Section B

Syllabus

- Definition and Properties of Continuous Time and Discrete Time Signals
- Types of Signals: Exponential, Sinusoidal, unit impulse, unit step etc.
- Definition and Properties of Systems
- Linear Time-Invariant Systems and their Properties
- Convolution of CT and DT LTI systems
- Fourier Series (excluding Discrete-Time)
- Fourier Transform (Both Continuous and Discrete, excluding FFT)

Question Pattern

- There will be **four sets of questions in total**. **One set** will contain **45 marks** and it will be **mandatory to answer**. The other **three sets** will contain **30 marks** each. You need to answer any **two** out of **these three sets**. The questions will be mathematical in nature.

- Marks distribution
    - Q1 (45 Marks): Mandatory
        - Properties of Signals, Even-odd decomposition of Signals, Graph plotting: 15
        - Fourier Transform (excluding FFT) and its properties: 20
        - Fourier Series: 5
        - Sinusoidal and exponential signals: 5
    - Q2 (30 Marks)
        - Fourier Transform (excluding FFT) and its properties
    - Q3 (30 Marks)
        - Convolution of LTI Systems
        - Properties of LTI systems
    - Q4 (30 Marks)
        - Properties of Signals, Even-odd decomposition of Signals
        - Fourier Series

Text Book

- Signals and Systems, Second Edition, by Alan V. Oppenheim et al.

MIT6.003 Course Page

https://ocw.mit.edu/courses/6-003-signals-and-systems-fall-2011/

Princeton ELE301 Course Page

https://www.princeton.edu/~cuff/ele301/

Suggestions

The suggestions provided below are only for practising mathematical problems. Please ensure you have a very good understanding of the theory before delving into these problems. **You should follow the textbook, the slides and any additional resources provided in the class** to understand the theoretical concepts.

You should solve all the class test questions for practice. They have been uploaded in the course page in moodle. In addition, please solve all the examples and exercises provided in all the slides taught in your class.

NB: Solutions to MIT6.003 quiz, hw etc are provided in the course page.

- **Properties and Types of Signals (Time scaling, shifting, upsampling, downsampling, Even-odd decomposition etc) *****
    - Textbook exercises: Chapter 1 -> 1.21(a-d), 1.22(a-d), 1.23, 1.25(a-c), 1.26(a-b)
    - ELE301 hw1: 5, 6
    - You should be able to draw graphs of any function of type **`f(x)= mx+c`** where **`m`** and **`c`** are constants. You should also be able to write down the definition of a function given its graphical plot(s). Please refer to CT 1 question.

- **Linear Time-Invariant Systems and their Properties ****
    - Textbook exercises: Chapter 2 -> 2.28, 2.29

- **Convolution of LTI Systems ****
    - Textbook exercises: Chapter 2 -> 2.2, 2.3, 2.6, 2.11, 2.13, 2.25, 2.26
    - MIT6.003 Quiz 2 Fall 2011: 3
    - MIT6.003 Quiz 2 Spring 2010: 1

- **Fourier Series ***
    - Textbook exercises: Chapter 3 -> 3.1, 3.3, 3.4, 3.6, 3.22(a-c), 3.26(a-b)
    - MIT6.003
        - Fall 2011 HW 8: 1, 2

- **Fourier Transform and its properties (Excluding FFT) *****
    - Textbook exercises: 4.1 - 4.4, 4.6, 4.21, 4.22, 5.21, 5.22
    - MIT6.003
        - Fall 2011 HW 9: 1, 2, 3
        - Fall 2011 HW 10: 4, 5
        - Fall 2011 Quiz 3: 1
        - Spring 2010 Final: 5

- **General Suggestions**
    - Solve all the class test questions.
    - Solve all the examples and exercises given in the slides taught by every course teacher.
    - Remember the necessary trigonometric and integration formula.